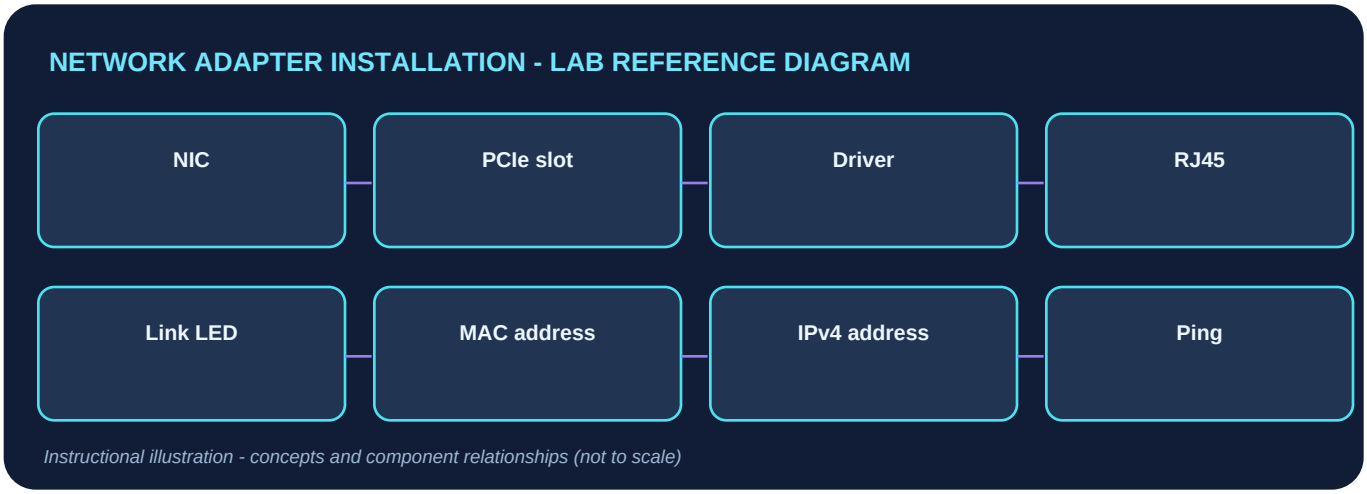
5.2.2 Select and Install a Network Adapter

STUDENT STUDY GUIDE + VOCABULARY + VISUAL REFERENCE

Learning goal

Install and test a NIC.

Reference image



Use the labeled illustration to identify the key components, commands, or checkpoints you will encounter in this lab.

Prepare for the simulation

1. Identify compatible NIC and correct expansion slot.
2. Install and attach Ethernet cable to switch.
3. Confirm OS detects NIC and correct driver.
4. Verify address, default gateway and ping connectivity.

Troubleshooting and safe practice

• No link: check cable, switch port and NIC LEDs.

• 169.254.x.x: investigate DHCP connectivity.

5.2.2 | VOCABULARY & REVIEW

Essential vocabulary

Term	Meaning / recognition cue
NIC	Network interface controller that connects a device to a network.
PCIe slot	Physical interface or connection point; verify its intended function and compatibility before use.
Driver	Software enabling the operating system to communicate with hardware.
RJ45	Common name for the 8P8C Ethernet modular connector.
Link LED	A key term in network adapter installation; identify its purpose, where it is used, and how to verify it.
MAC address	Memory characteristic or configuration item; check motherboard support and installation requirements.
IPv4 address	Memory characteristic or configuration item; check motherboard support and installation requirements.
Ping	A key term in network adapter installation; identify its purpose, where it is used, and how to verify it.

Knowledge check - answer in your own words

1. Identify two components or commands in the reference diagram and explain what each does.

2. What should you verify before changing or installing hardware, settings, or files in this lab?

3. Choose one listed failure symptom. What is your first safe diagnostic step?

4. How will you prove the task is complete, rather than merely assuming success?

Lab readiness checklist

[ ] I can identify the labeled reference items. [ ] I understand the safety precautions.

[ ] I can explain the vocabulary. [ ] I can verify and document successful results.

Student: _____ Date: _____